

Problem 4

A wheel of radius $R = 0.5$ m is rolling along a smooth horizontal path without slipping. The velocity of the wheel's center is $c = 2$ m/s. Calculate the linear velocities of the points on the circumference of the wheel that are displaced by an angle $\pi/2$ radians relative to the contact point of the wheel with the ground.

Solution

Given that the wheel is rolling without slipping, the velocity of the contact point with the ground is zero. The linear velocity of the center of the wheel is:

$$v_c = c = 2 \text{ m/s.}$$

The angular velocity of the wheel is given by:

$$\omega = \frac{v_c}{R} = \frac{2}{0.5} = 4 \text{ rad/s.}$$

We are asked to find the linear velocities of the points on the circumference of the wheel displaced by an angle $\pi/2$ radians from the contact point with the ground. These points are as follows:

1. Point at the top of the wheel (i.e., $\pi/2$ radians above the contact point).
2. Point at the front of the wheel (i.e., $\pi/2$ radians forward from the contact point).

Point at the Top of the Wheel

At the top of the wheel, the velocity due to rotation is in the same direction as the translational velocity. The linear velocity at this point is:

$$v_{\text{top}} = v_c + \omega R = 2 + 4 \times 0.5 = 2 + 2 = 4 \text{ m/s.}$$

Point at the Front of the Wheel

The velocity at the front of the wheel is perpendicular to the translational velocity. The velocity due to rotation is directed upward. The resultant velocity is the vector sum of the translational velocity and the velocity due to rotation. The magnitude of the velocity is:

$$v_{\text{front}} = \sqrt{v_c^2 + (\omega R)^2}.$$

Substituting the known values:

$$v_{\text{front}} = \sqrt{2^2 + (4 \times 0.5)^2} = \sqrt{4 + 2^2} = \sqrt{4 + 4} = \sqrt{8} \approx 2.83 \text{ m/s.}$$

Final Answer

- Velocity at the top of the wheel: $v_{\text{top}} = 4$ m/s.
- Velocity at the front of the wheel: $v_{\text{front}} \approx 2.83$ m/s.